

Project Design Phase-II Technology Stack (Architecture & Stack)

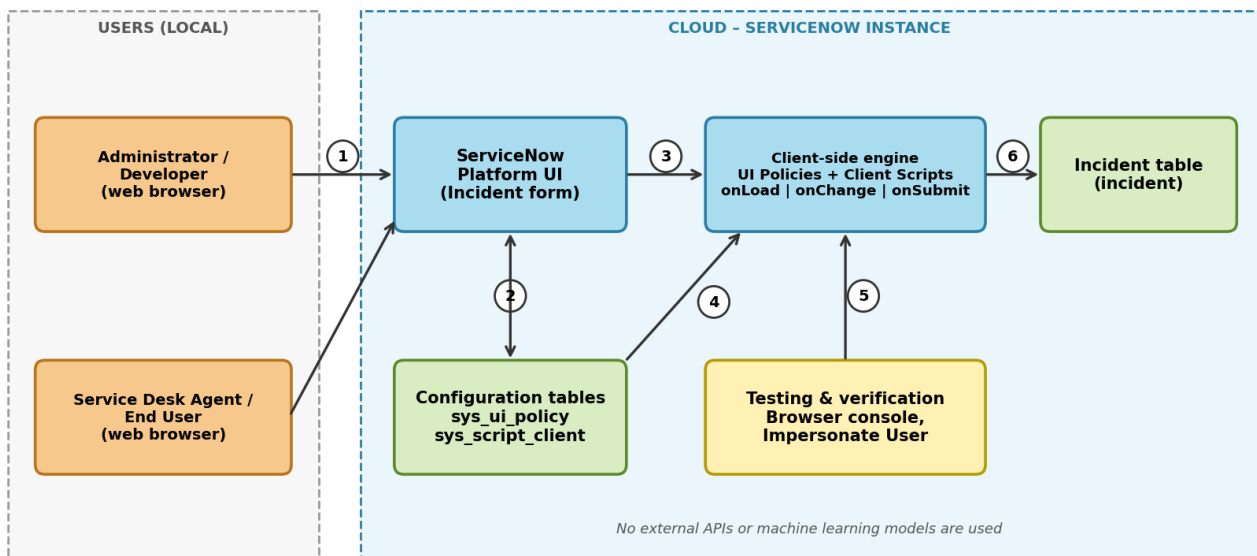
Date	03-10-2026
Team ID	SWTID-2026-8713
Project Name	Implement Client Script & UI Policy (Incident)
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2.

Architecture: Client Scripts and UI Policies on ServiceNow

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Flow

1. The administrator, developer and agents open the ServiceNow Platform UI in a web browser.
2. The developer creates UI Policies and Client Scripts, which are stored in the configuration tables (sys_ui_policy, sys_script_client).
3. When the Incident form is opened, the form and its rules are delivered to the browser.
4. The client-side engine runs the onLoad Client Scripts and UI Policies to set the initial form state.
5. When a field changes, the onChange scripts and UI Policy conditions update the form without submitting it.
6. On submit, the onSubmit script validates the data and only valid records are saved to the Incident table.
7. Testers check results with the form, the browser console and Impersonate User.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How the administrator, developer and agent interact with the application: the Incident form in the ServiceNow Platform UI in a web browser.	ServiceNow Platform UI (HTML, CSS, JavaScript in the browser)

S.No	Component	Description	Technology
2.	Application Logic-1	UI Policies with conditions and actions that make fields mandatory, visible or read-only.	ServiceNow UI Policy (client-side)
3.	Application Logic-2	Client Scripts that run onLoad, onChange and onSubmit and control the form using g_form and g_user.	Client-side JavaScript (ServiceNow Client Scripts)
4.	Application Logic-3	UI Policy scripts (Execute if true / Execute if false) and the order in which policies and scripts run.	ServiceNow UI Policy scripts
5.	Database	Incident table (incident) and the tables that store the rules (sys_ui_policy, sys_script_client).	ServiceNow platform database
6.	Cloud Database	Database service on the cloud: records and rules are stored in the ServiceNow instance.	ServiceNow cloud instance
7.	File Storage	No separate file storage is required; the project works with form rules and records only.	Not applicable
8.	External API-1	No external API is used in this solution.	Not applicable
9.	External API-2	No external API is used in this solution.	Not applicable
10.	Machine Learning Model	No machine learning model is used in this solution.	Not applicable
11.	Infrastructure (Server / Cloud)	Application deployment on cloud. Local Server Configuration: not applicable, no local server is used. Cloud Server Configuration: ServiceNow-hosted instance.	Cloud (ServiceNow instance)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No open-source framework is added; the solution uses native ServiceNow client-side scripting.	ServiceNow client-side JavaScript
2.	Security Implementations	Role-based access with server-side ACLs for enforcement; client rules guide data entry; testing with Impersonate User and the browser console.	ServiceNow roles, ACLs, Impersonate User
3.	Scalable Architecture	The same UI Policy and Client Script patterns can be applied to other forms; the platform separates the browser, application and database tiers.	ServiceNow multi-tier cloud platform
4.	Availability	Rules are delivered from the cloud-hosted ServiceNow instance, so they apply whenever the instance is available.	ServiceNow cloud instance
5.	Performance	Scripts stay short, run in the browser and avoid synchronous server calls (for example, GlideRecord) on the form.	ServiceNow Client Scripts and UI Policies

References:

<https://developer.servicenow.com>
<https://docs.servicenow.com>
<https://c4model.com/>